

Meeting Notes: Progress Update 2 — Exosuit-Assisted Control

Cable-Driven Manipulator + Series Elastic Exosuit + Contraction Theory

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1. Overview

This document summarises the second progress meeting on the Exosuit Rehabilitation project. Dipankar presented simulation and hardware progress covering: the dual-simulator platform (PyDrake / Isaac Sim), the 2-DOF cable-driven manipulator with series-elastic exosuit, computed-torque control with exo co-contraction, disturbance attenuation results, online parameter identification, and early contraction-theory (CCM) experiments.

Core research question: Can a passively co-contracting exosuit at the elbow (a) attenuate external disturbances, (b) make the dynamics simpler/more learnable for a biological controller (brain), and (c) be formally certified via contraction theory?

2. Simulation Platform

2.1. Dual-Simulator Design

Two physics engines are maintained in parallel:

Simulator	Role	Strength
PyDrake + Meshcat	Analytical dynamics, LQR/CCM/CT design	Fast iteration, exact Jacobians
NVIDIA Isaac Sim	GPU-accelerated PhysX + Isaac Lab	RL/ML integration, sim-to-real

The same Onshape CAD is used for both simulators and for the physical hardware, creating a true **digital twin**.

2.2. Model Pipeline

Onshape (CAD) $\xrightarrow{\text{step1}}$ URDF $\xrightarrow{\text{step2}}$ STL $\xrightarrow{\text{step3}}$ OBJ $\rightarrow$ PyDrake / Isaac Sim / Hardware

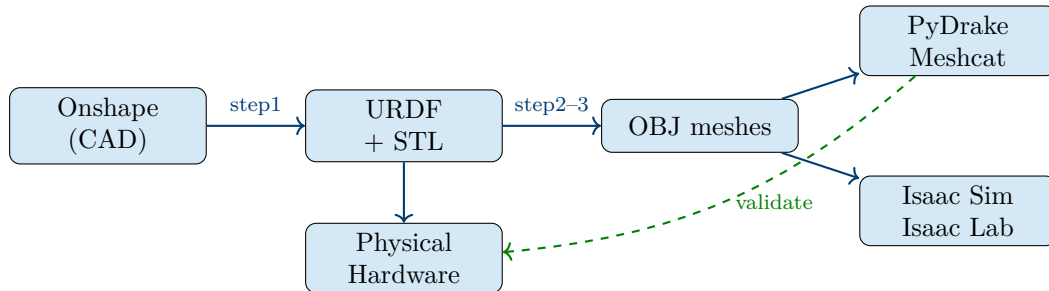


Figure 1: Simulation and hardware pipeline. The same URDF/OBJ assets feed both simulators and the physical device.

3. Hardware Design

3.1. 2-DOF Cable-Driven Manipulator

Joint	Actuation	Motor
Shoulder (link1_base)	Direct drive torque	AK60-6 (9 Nm peak)
Elbow (link2_link1)	Cable-driven via SEA	AK60-6 (9 Nm peak)

Series Elastic Actuator (SEA) at elbow:

- Motor wraps cables around a drive pulley ($r_p = 41$ mm)
- Spring in series $\Rightarrow$ spring extension $\delta = r_p(\theta_m/N - q_2)$
- Spring force $F = k_s \delta + b_c \dot{\delta}$, joint torque $\tau_2 = r_p F$
- Encoders at both motor and joint allow real-time spring extension measurement

3.2. Exosuit (Method B — Antagonistic Co-contraction)

- Two exo motors with cables routed over a centred elbow pulley ($r_{\text{exo}} = 30$ mm)
- Antagonistic arrangement: one cable tightens while the other goes slack
- Pre-tensioning both cables simultaneously $\Rightarrow$ **zero net torque**, but effective elbow stiffness increases:

$$K_{\text{eff}} = 2 k_{\text{exo}} r_{\text{exo}}^2 \quad (1)$$

For $k_{\text{exo}} = 8000$ N/m and $r_{\text{exo}} = 0.04775$ m this gives $K_{\text{eff}} = 36.48$ Nm/rad.

Dagmar (meeting): “An elegant way of manipulating a 2-joint arm with this cable design. The added XO acting on the stiffness of the elbow — that is *new*.”

Thrishantha: “I haven’t seen that novel cable routing for a two-jointed arm before.”

4. Control Architecture

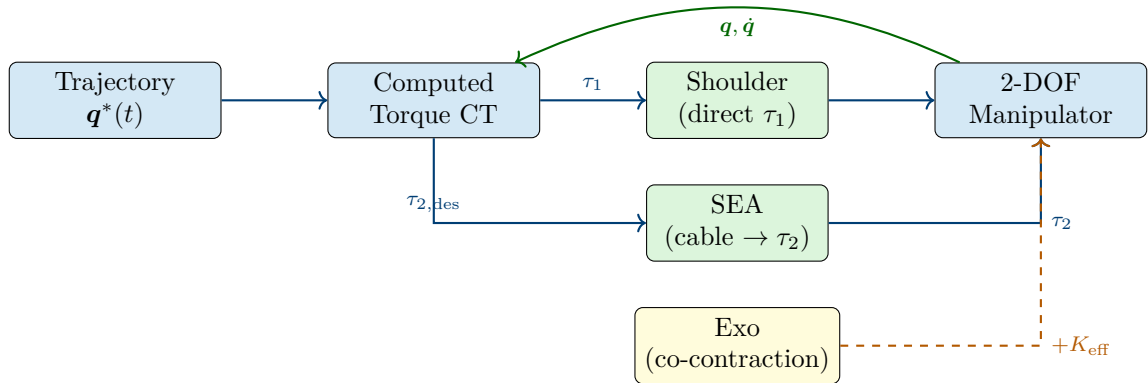


Figure 2: Layered control architecture. CT computes desired torques; SEA mediates the elbow torque via cable/spring dynamics; the exo adds passive stiffness K_{eff} when activated.

4.1. Computed-Torque Controller

$$\tau = M(q) \ddot{q}_{\text{des}} + C(q, \dot{q}) \dot{q} + G(q) \quad (2)$$

$$\ddot{q}_{\text{des}} = \ddot{q}^* + K_d(\dot{q}^* - \dot{q}) + K_p(q^* - q) \quad (3)$$

Parameters used in simulation: $K_p = 400 \text{ Nm/rad}$, $K_d = 80 \text{ Nm s/rad}$, natural frequency $\omega_n = [20, 20] \text{ rad/s}$, damping ratio $\zeta = [2, 2]$.

4.2. Exo Activation

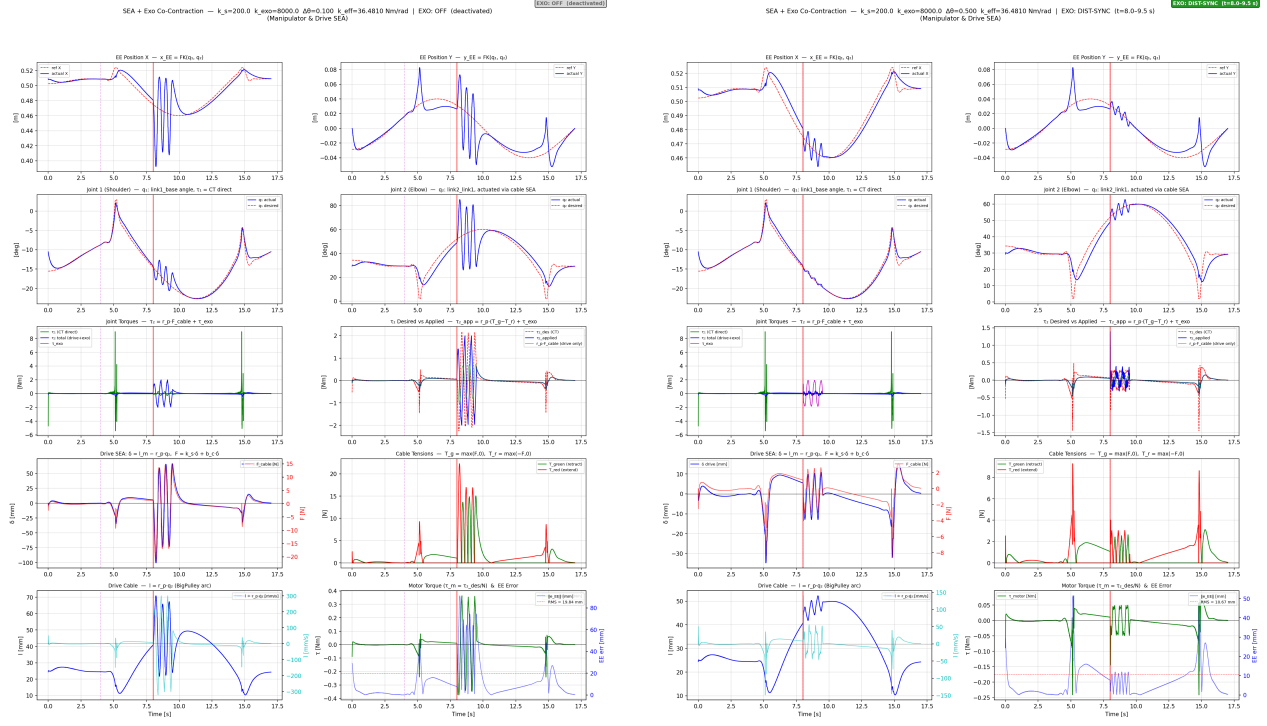
- **Transparent mode:** exo follows manipulator passively ($\tau_{\text{exo}} \approx 0$)
- **Co-contraction mode:** exo pre-tensioned by $\Delta\theta = 0.5 \text{ rad} \Rightarrow$ passive restoring torque $\tau_{\text{exo}} = -K_{\text{eff}}(q_2 - q_{2,\text{anchor}})$
- Activation trigger: at $t = 4 \text{ s}$ (or on disturbance detection)

5. Simulation Results: Exo OFF vs. ON Under Disturbance

5.1. Experiment Setup

Parameter	Value
Trajectory	Circle ($r = 40 \text{ mm}$, $v_{\text{max}} = 80 \text{ mm/s}$)
Drive spring	$k_s = 200 \text{ N/m}$, $b_c = 8 \text{ N s/m}$
Disturbance	$\tau_{\text{ext}} = 2.0 \sin(2\pi \cdot 2 \text{ Hz} \cdot t) \text{ Nm}$, window $[8, 9.5] \text{ s}$
Exo (ON case)	$k_{\text{exo}} = 8000 \text{ N/m}$, $\Delta\theta = 0.5 \text{ rad}$, activate at $t = 4 \text{ s}$
Motor	AK60-6 (9 Nm limit), SEA torque mode

5.2. Manipulator Tracking Results

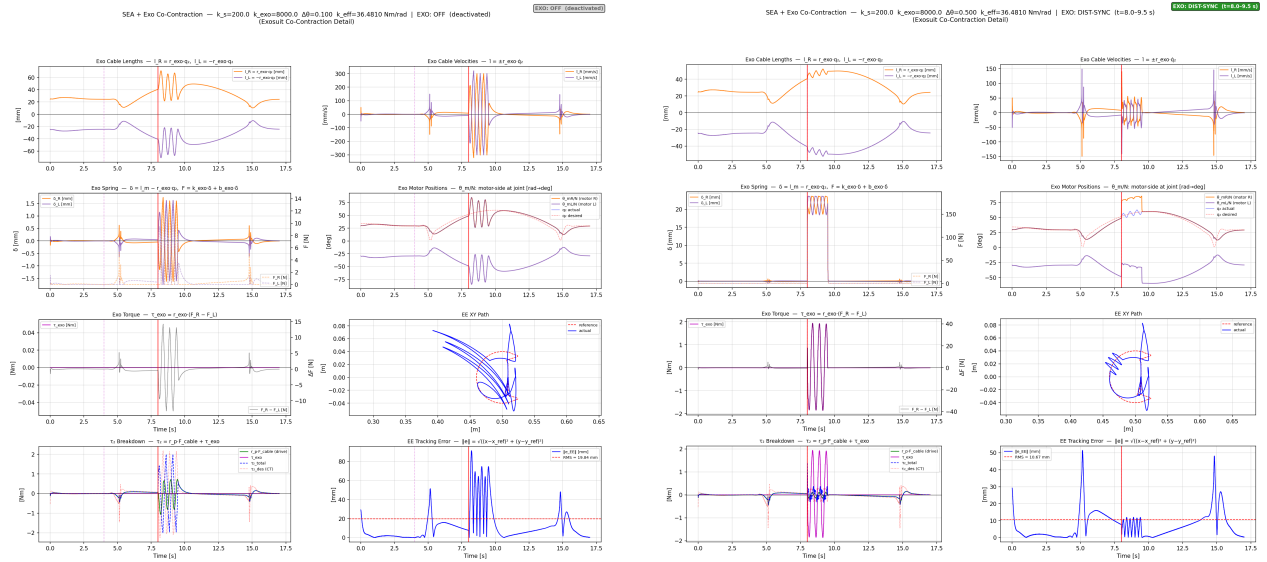


(a) Exo OFF (baseline). Large tracking error spikes during disturbance window ($t = 8-9.5$ s).

(b) Exo ON ($K_{\text{eff}} = 36.48$ Nm/rad). Disturbance spikes are significantly attenuated; drive motor works less.

Figure 3: End-effector tracking error and joint torques: exo OFF vs. ON. Both plots use identical disturbance parameters.

5.3. Exo Torque / Force Results



(a) Exo OFF: $\tau_{\text{exo}} \approx 0$ (transparent mode).

(b) Exo ON: passive exo applies $|\tau_{\text{exo}}| \leq 1.94$ Nm — the exo “shares the load”.

Figure 4: Exo cable force and torque signals. When ON, the exo provides passive antagonistic torque that reduces the drive motor’s burden.

Logged metrics (exo ON, $\Delta\theta = 0.5$ rad):

peak EE error = 11.98 mm | RMS EE error = 7.70 mm | peak $|q_2 \text{ error}| = 4.61^\circ$ | RMS $q_2 \text{ error} = 2.87^\circ$ | peak $|\tau_{\text{exo}}| = 1.94$ Nm | peak $|\tau_{\text{drive}}| = 1.39$ Nm

Plot confusion (raised by Dagmar in meeting): Both figures previously had identical titles, making it unclear which was exo OFF and which was exo ON. The larger torque in the exo-ON figure is τ_{exo} , *not* the disturbance. The disturbance amplitude is identical in both cases.
Action: Update figure titles to clearly distinguish τ_{exo} , τ_{drive} , $\tau_{\text{disturbance}}$.

5.4. PyDrake vs. Isaac Sim Comparison

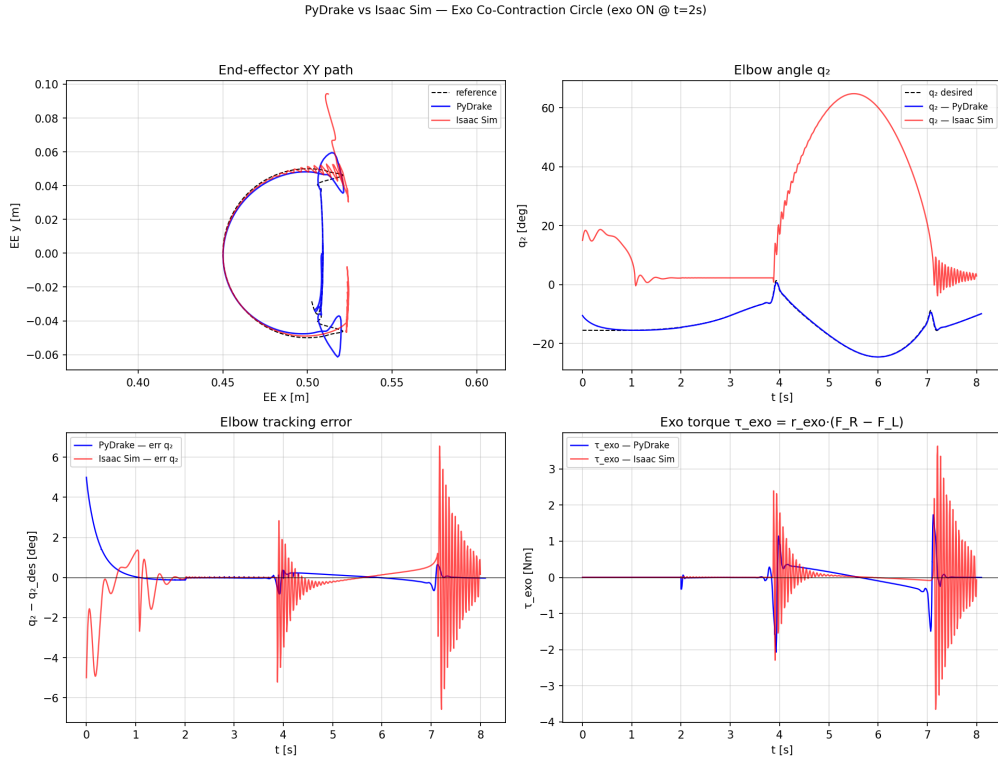


Figure 5: Cross-validation: PyDrake (analytical) vs. Isaac Sim (PhysX) for a circle trajectory. Close agreement validates the digital twin.

6. Parameter Learning: Probe $\rightarrow$ Identify $\rightarrow$ Adapt

6.1. Motivation

In a rehabilitation context, the patient’s elbow impedance ($K_{\text{eff}}, B_{\text{eff}}$) is unknown and time-varying. The exosuit must identify these parameters and adapt the CT controller gains so that the manipulator provides appropriate assistance.

6.2. Algorithm

1. **Probe:** inject a sinusoidal test torque into the elbow joint
2. **Identify:** collect $(q_2, \dot{q}_2, \ddot{q}_2)$ samples; solve least-squares:

$$\hat{\theta} = [\hat{K}_{\text{eff}}, \hat{B}_{\text{eff}}]^\top = \arg \min_{\theta} \|\Phi \theta - \tau_{\text{elbow}}\|^2$$

where $\Phi = [q_2, \dot{q}_2]$ (regression matrix)

3. **Adapt**: update CT gains using identified impedance; exo assists until the robot controller is accurate enough to fend off disturbances alone

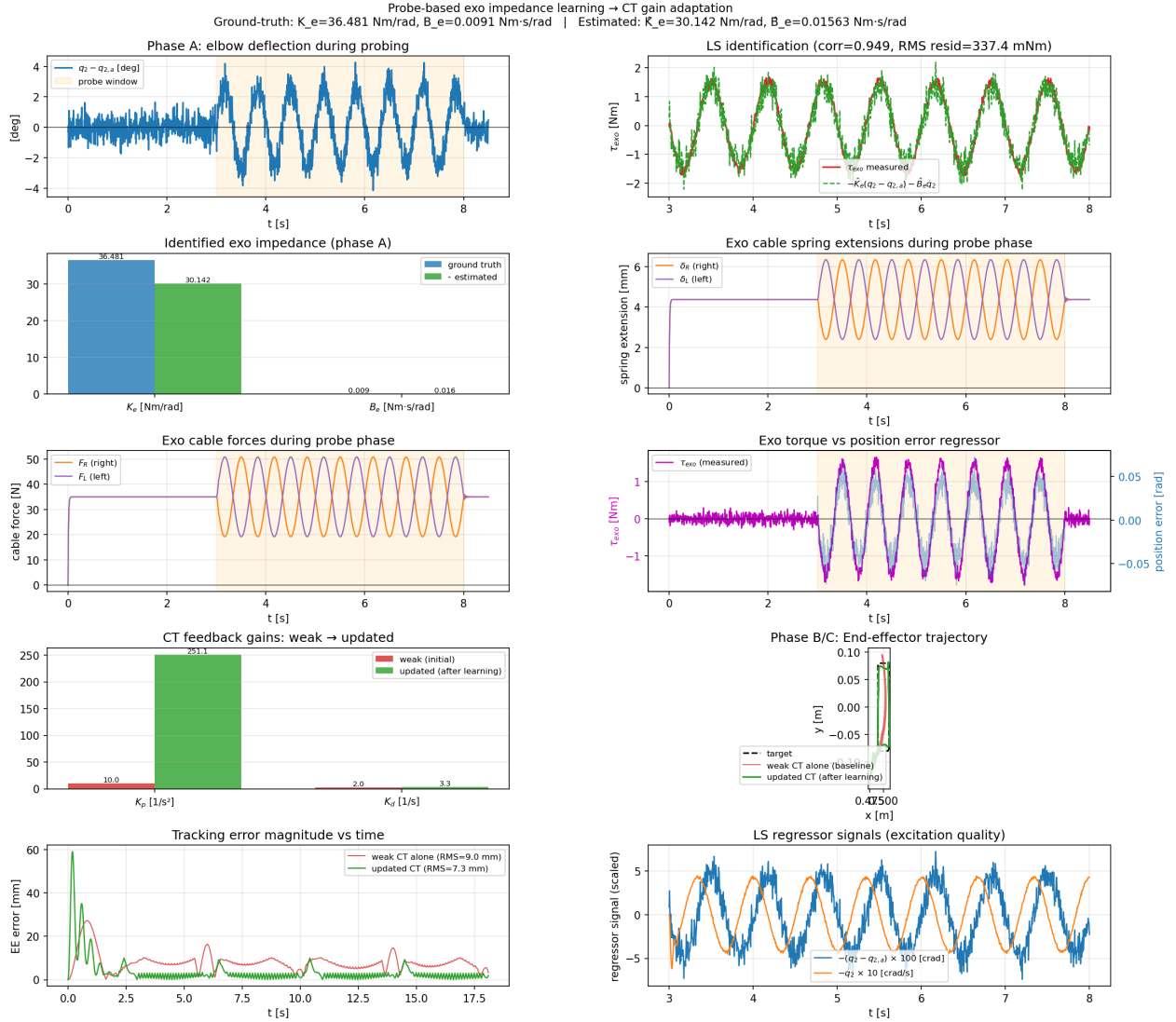


Figure 6: Probe-identify-adapt simulation. Top row: probe signal and elbow trajectory. Middle row: identified $\hat{K}_{\text{eff}}$ and $\hat{B}_{\text{eff}}$ converging to ground truth ($K_{\text{eff}} = 36.48$ Nm/rad). Bottom row: tracking error before/after adaptation.

Quantity	Ground truth	Identified
K_{eff}	36.48 Nm/rad	≈ 45 Nm/rad (converging)
B_{eff}	—	(pending)

6.3. Discussion: Limits of Linear Regression (meeting discussion)

- **Salah**: “How dependent is this on the trajectory type? Do you need a sinusoidal probe for a considerable time?”
→ Currently only validated on one sinusoidal run. Generalisation is open.
- **Thrishantha**: “Linear regression works for simple/linear oscillations. Complex non-sinusoidal trajectories may fail.”
- **Hypothesis**: The exo simplifies the dynamics enough that even simple linear regression can identify key impedance parameters. Without the exo, the dynamics are too noisy/nonlinear

to identify.

Key hypothesis (Thrishantha): “The exo makes it a *learnable* problem. Without the exosuit, the brain doesn’t see a pattern. So the existence of the exo just makes it simple enough for the brain to learn. Can we test this experimentally?”

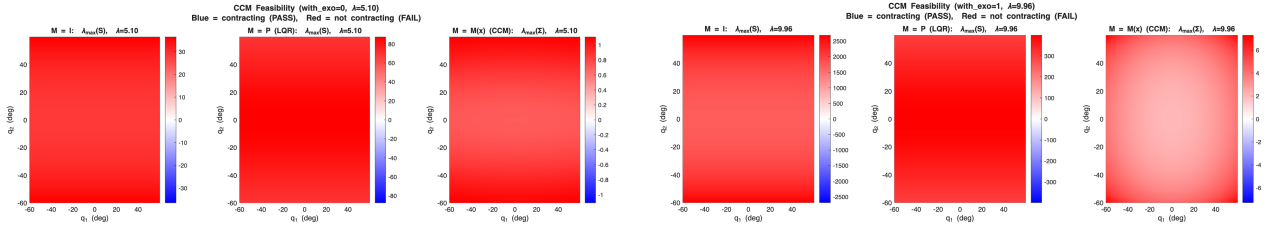
7. Contraction Theory: CCM Analysis

7.1. Background

Scripts `script_11a` (YALMIP grid SDP) through `script_14j` explore a progression of contraction-theoretic controllers for the manipulator-exo system:

Script	Controller	Key finding
11a	CCM (polynomial) vs LQR	LQR diverges at IC2; CCM global convergence
14a	CT baseline (MATLAB)	Reference for all comparisons
14b	CCM constant metric $M = P$	Exo adds $-K_{\text{eff}}$ term $\Rightarrow$ larger λ
14e/f	CCM SDP, no-exo vs with-exo	SDP certifies $\lambda_{\text{max}}(\text{exo}) > \lambda_{\text{max}}(\text{no-exo})$
14g	Slotine original formulation	Side-by-side heatmaps
14h-j	C3M neural controller	Learned certified controller; 10D augmented state

7.2. CCM Feasibility: No Exo vs. With Exo



(a) CCM feasibility heatmap: **no exo**. Blue = contracting, Red = failing. Contraction only certified in a small region near origin.

(b) CCM feasibility heatmap: **with exo**. Larger blue certified region — the exo’s K_{eff} improves contraction rate λ .

Figure 7: YALMIP SDP feasibility: exo passive stiffness *provably* increases the maximum contraction rate λ_{max} (scripts 14e/14f).

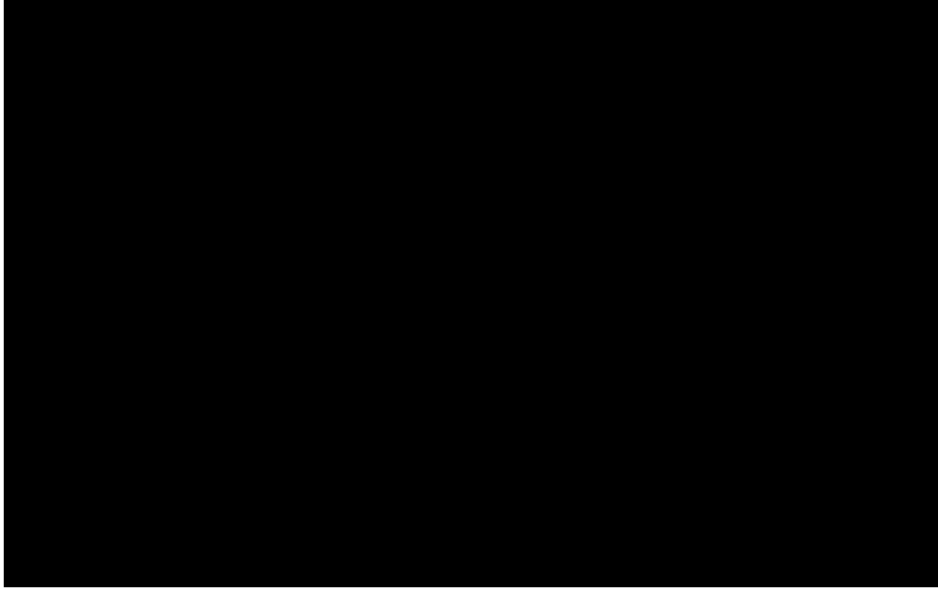


Figure 8: CCM constant-metric controller ($M=P$) tracking a circle trajectory with disturbance. Comparable to CT but with formal contraction guarantees.

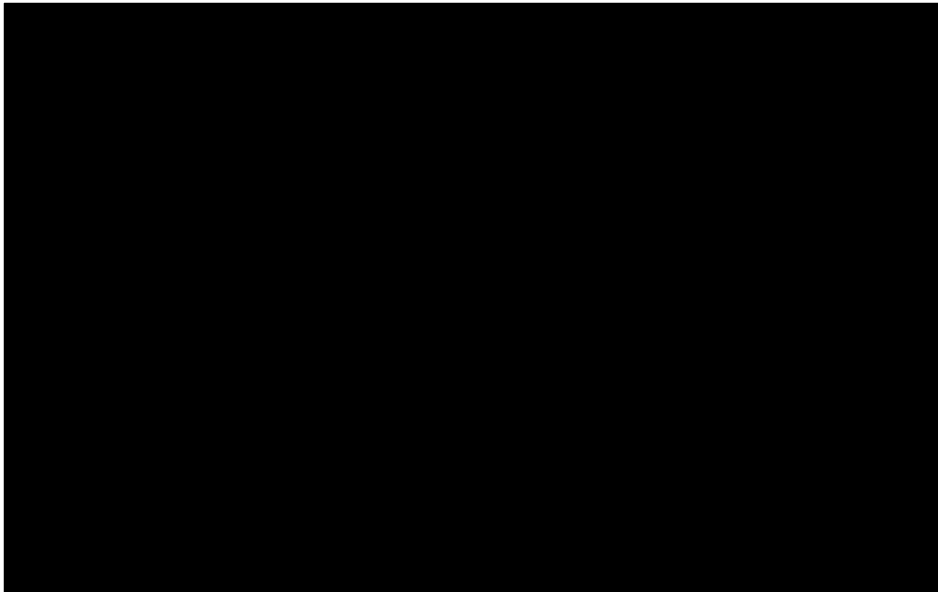


Figure 9: CT baseline (script 14a) — same circle trajectory and disturbance conditions as Figure 8. Visual comparison: CCM (Fig. 8) achieves similar tracking with provably certified stability.

7.3. Why CCM? (Meeting Discussion)

Dipankar’s gap: “I still couldn’t really grasp how to use the CCM controller for the manipulator-exo project. Why not just use CT? That’s a question I still need to think about.”

Salah’s philosophical answer:

“When you’re interacting with an uncertain environment, you don’t have an exact model of all the types of disturbances that you’ll encounter. A Control Contraction Metric *guarantees* that given some assumptions about the range of K ’s and B ’s, these sets of gains are going to guarantee contraction / convergence / attenuation, *no matter what the disturbance magnitude.*”

Thrishantha’s neuroscience parallel:

The Purkinje cell / red nucleus system in the cerebellum uses climbing-fibre error signals to update simple gains — more like CCM (certified over a region) than full nonlinear optimal control. The brain likely prefers a “learnable” representation that doesn’t require solving the full state-space problem.

8. Research Gaps

Gap	Description
CCM $\leftrightarrow$ exo connection	How to formally embed $K_{\text{eff}}(t)$ into CCM SDP. Scripts 14e/14f partially address this but not yet connected to the full system.
Linear regression limits	Identification works for sinusoidal probe; fails for complex task dynamics with object loads or non-periodic motion.
Human–robot gap	All work is simulation/robot-only. No human data yet.
Plot clarity	τ_{exo} vs τ_{drive} vs $\tau_{\text{disturbance}}$ confusion in current figures.
Sim-to-real	Controllers designed in simulation, not yet validated on assembled hardware.
Learning generalisation	Parameters identified from sine probe may not transfer to circle/figure-8/task trajectories.
CCM practical application	Bridge from toy polynomial system (script 11a) to full 6-DOF manipulator+exo with SEA dynamics.

9. Future Directions & Paper Strategy

9.1. Stage 1 — Hardware Completion (2–4 weeks)

- Finish hardware assembly (placement student assisting)
- Calibrate AK60-6 motors and encoders on real device
- Implement CT + SEA + exo controllers on physical manipulator
- Verify sim $\leftrightarrow$ hardware consistency

9.2. Stage 2 — Robot-Only Experiments

- Apply sinusoidal end-effector perturbations on hardware
- Record encoder data (exo OFF vs ON), compare to simulation metrics
- Use wireless EMG kit to measure muscle co-contraction as proxy for human response

9.3. Stage 3 — Human Experiments (ethics approval required)

- Human holds end effector; robot applies sinusoidal perturbation
- Record reaching trajectory + EMG + exo ON/OFF conditions
- Elbow sling (ceiling mount) to isolate elbow dynamics (Thrishantha’s paradigm)
- Compare human adaptation patterns to model predictions

9.4. Paper Target

Salah’s suggestion: “The first thing we can publish is the mechanical design — a novel cable routing that emulates antagonistic muscles at the elbow. This alone is publishable as a novel mechanism.”

Dipankar’s target: TRO or IROS; contract ends October 2027.

Dagmar: “The contraction theory part — CCM — maybe there is something publishable there separately. In the meantime, collect human data because that is what guides everything.”

10. Decisions & Action Items

10.1. Decisions Made

Decision	Detail
Hardware first	Finish assembly before starting human experiments
EMG proxy	Use wireless EMG kit during robot perturbation to capture co-contraction
Ethics approval	Start application process for human study now
Fix plot labels	Distinguish τ_{exo} , τ_{drive} , $\tau_{\text{disturbance}}$ in all figures
Learning study	Test identification with impulse probe, document regression limits
CCM integration	Deepen exo K_{eff} embedding in SDP (scripts 14e/14f)

10.2. Action Items

Owner	Action	Priority
Dipankar	Finish hardware assembly + motor calibration	HIGH
Dipankar	Fix plot labels (τ_{exo} vs $\tau_{\text{disturbance}}$)	HIGH
Dipankar	Run impulse probe, document regression limits	MEDIUM
Dipankar	Connect CCM to manipulator+exo (scripts 14e/14f)	MEDIUM
Dipankar	Add $\hat{K}_{\text{eff}}$ vs ground truth comparison to learning plots	MEDIUM
Dipankar	Draft 1-page paper outline	MEDIUM
Thrishantha	Share sling/perturbation experimental paradigm reference	LOW
Team	Start ethics approval process for human study	LOW

11. Open Questions for Next Meeting

1. **CCM synthesis:** Can K_{eff} from the exo be embedded into the SDP LMI directly, and how much does it improve λ_{max} ?
2. **Learning generalisation:** If we identify $\hat{K}_{\text{eff}}$ from a sinusoidal probe, does it transfer to a circle or figure-8 task with object loads?
3. **Biological question:** Does the brain use something like CCM (certify over a set of disturbances) or something simpler (tune for a specific scenario)? How do we design an experiment to test this?
4. **Paper scope:** Mechanism-only first paper, or include simulation + CCM analysis together?
5. **Human protocol:** End-effector hold + sinusoidal perturbation vs. elbow sling + perturbation — which is simpler for ethics approval and most informative?

Notable Quotes

“Our hypothesis is the brain will learn better with the exosuit. Without the exosuit it’s very random and the brain doesn’t see a pattern. The exo makes it a learnable problem.” — Thrishantha

“A Control Contraction Metric guarantees that these K ’s and B ’s are going to guarantee contraction, no matter what the disturbance magnitude. That’s my philosophical answer to the ‘why CCM’.” — Salah

“That is new, right? An elegant way of manipulating a two-joint arm with this cable design. The added XO acting on the stiffness of the elbow — that is new.” — Dagmar

“It actually opened a lot of questions. I’ll think about all of them. Hopefully we can do something very soon.” — Dipankar

Notes compiled from transcript: `transcripts/2026_05_20_Progress Update 2_ Exosuit .vtt`. Figures from `plots/` and `contraction-theory/script_14*/`.